MATHEMATICAL MODEL OF THE COOLANT FLOW IN CYLINDRICAL TUBES IN THE AUTOMOBILES RADIATOR

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DECLARATION

I'm **TALHA SAJID** Roll No. **0291-MPHIL-MATH-21** student of MPhil at **Government College University Lahore** in the subject of Mathematics session (2021-2023), hereby declare that the matter printed in this thesis titled **"Mathematical Model of the Coolant Flow in Cylindrical Tubes in the Automobiles Radiator"** is my own work and I have not taken any material from any source except referred to wherever due that amount of plagiarism is within the acceptable range.

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(TALHA SAJID)

<u>COMPLETION CERTIFICATE</u>

Certified that the work contained in this dissertation titled **"Mathematical Model of the Coolant Flow in Cylindrical Tubes in the Automobiles Radiator"** has been carried out and completed by **TALHA SAJID**, Roll No.**0291- MPHIL-MATH-21** under my supervision during his studies for MPhil in Mathematics.

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APPROVAL

This thesis titled **"Mathematical Model of the Coolant Flow in Cylindrical Tubes in the Automobiles Radiator"** by **TALHA SAJID**, Roll No.**0291-MPHIL-MATH-21** fulfills the required standards and has been approved for the award of the degree of Master of Philosophy in Mathematics.

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<u>DEDICATION</u>

This thesis is dedicated to my beloved ***father and late mother***. Your unwavering love, support, and encouragement have been the guiding lights of my academic journey. Without your sacrifices and belief in me, this achievement would not have been possible.

This thesis represents not only my dedication to my field of study but also a testament to the love and support that you have showered upon me throughout my academic journey. Thank you for being my pillars of strength, my biggest cheerleaders, and my role models.

With all my love and gratitude,

Talha Sajid

Abstract

In this thesis, we will discuss the coolant flows via cylindrical tubes in an automotive radiator. The mathematical model of the radiator system in the form of couples differential equations governing temperature, concentration, and coolant velocity are employed. The solution of the model is obtained by using the Laplace and finite Hankel transforms.The coolant flow velocity, temperature, and concentration analytical expressions are obtained for flow of coolant in radiator. The effects of the order of radiation parameter, metabolic heat source, Peclet number, and other significant parameters are considered, and valuable outcomes are summarized using numerical simulations and graphical demonstrations to validate the arguments for the better choice of coolant and the diameter of the radiator system tubes.

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In the name of Allah, the Most Gracious and the Most Merciful

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Contents

Chapter 1

Preliminaries

1.1 Fluids

Definition 1.1.1. A *fluid*, which can include both liquids and gases, is a substance that can flow and take on the shape of its container. Fluids are categorized as real fluid, Ideal fluid, Newtonian and Non-Newtonian fluid.

1.1.1 Properties of fluids

Definition 1.1.2. The property of a fluid to resist distortion is called *viscosity*. It is conceivable that a force of friction will develop between fluid layers that are next to one another when the layer with lower velocity attempts to slow the layer with higher velocity. There are no fluids whose measured viscosity is zero because all types of fluid flow occupy viscous properties to varying degrees. When working with fluids, different fluids have varied viscosities; for example, as temperature rises, liquids tend to have less viscosity whereas gases have more viscosity.

Definition 1.1.3. *Viscoelasticity* is the behavior of a material when it experiences deformational processes while exhibiting both viscosity and elasticity i.e., Oils, blood, and lubricants etc.

Definition 1.1.4. $Newton's\ law\ of\ viscosity$ defines the relation between a fluid's internal velocity gradient and the shear stress that is applied to it.The mathematical representation of Newton's law of viscosity is given by the following equation:

$$\tau = -\mu \frac{du}{dy}$$

Where τ is shear stress applied to the fluid, μ is dynamic viscosity.

Definition 1.1.5. In the field of physics known as *magnetohydrodynamics* (MHD), magnetic particles and electrically conducting fluids are studied. Current is created by the magnetic field. Due to its applications and new studies in this area, MHD is quite significant.Architects employ MHD standards in the design of heat exchangers, siphons, and flow meters, as well as in the propulsion, control, and reentry systems of spacecraft, to develop cutting-edge force-generation frameworks.

1.1.2 Classification of fluids

Definition 1.1.6. $Viscous\ flow$ is the word used to describe the flow of fluids where viscosity dominates or can be seen in the flow patterns. This might occur in fluids with rather significant velocity gradients. Usually, flows near to cylinder walls are handled like viscous flows, much like the walls of pipes.

Definition 1.1.7. The terms "*inviscid*" and "$non-viscous$" refer to flows that don't have viscosity or have viscosity as a subordinate property. The movement of liquid or gases through the center of various cylindrical objects, such as pipes

Definition 1.1.8. A $Newtonian\ fluid$ adheres to Newton's law of viscosity, which asserts that the fluid's rate of distortion (shear rate) is directly linked to the applied shear stress.

Definition 1.1.9. $Non - Newtonian\ fluids$ are ones whose proportionality constants, such the viscosity co-efficient, vary with the rate of deformation. Blood, liquid plastic, and polymer are examples of non-Newtonian fluids.

Definition 1.1.10. The term "*compressible fluids*" refers to any fluid whose density changes as a result of an external force or pressure. They contain gases.

Definition 1.1.11. *Incompressible fluids* are those whose density is constant during the flow and does not change. Since liquids' densities are generally constant, it follows that their flow cannot be compressed.

1.1.3 Classification of fluid's flow

Definition 1.1.12. A *steady flow* is one in which the properties of the fluid, such as pressure, velocity, and density, are independent of any point's position in time. An example would be a steady flow from a pipe with variable diameter and constant pressure in any cylindrical tank. *Unsteady flow*, on the other hand, is the flow in which fluid properties like pressure, velocity, and density depend on time. For instance, movement of water in a river during a heavy rainstorm.

Definition 1.1.13. *Laminar flow* is a smooth flow pattern that occurs when particles moving through a liquid do not interact with one another. The flow path is predictable, and the particle velocity is consistent throughout. An illustration of laminar flow is coolant moving through the radiator tubes.
Turbulent flow is defined as a liquid flow where the liquid particle tracks are neither smooth or regular. The fluid fluctuates in an unpredictable manner. Fluids also have a constant rate of change that defies all laws of direction and magnitude. It displays significant pressure and velocity changes.A river or canal's high-velocity water flow is a representation of turbulent flow.

1.2 Automobiles Radiator

Definition 1.2.1. , The *radiator* serves as a heat exchanger. It typically consists of multiple small diameter pipes with fins attached and is constructed of aluminum. Furthermore, it facilitates the transfer of heat from the engine's hot water to the ambient air. It also includes an exit port, an inlet port, a securely sealed lid, and a drainage plug.

Definition 1.2.2. The *water pump* circulates the cooled coolant back to the cylinder block, heater core, and cylinder head. Eventually, the liquid makes its way back to the radiator for further cooling.

Definition 1.2.3. It functions as a *thermostat* with a cooling valve feature, which opens to allow coolant to flow into the radiator when a specific temperature threshold is met. Paraffin wax, which expands and opens at a specific temperature, is present in the thermostat. A thermostat is used by the cooling system to control the internal combustion engine's normal operating temperature. The thermostat is activated after the engine reaches normal operating temperature. The radiator can then receive the coolant.

Definition 1.2.4. This is a $freeze\ plug$ that is used to plug gaps left behind from the casting process in the cylinder block and cylinder heads. If there is no frost protection, they may pop out in cold weather.

Definition 1.2.5. The radiator's largest component is its *core*. It is a metal block with metal cooling fins that aid in air venting. Hot liquid releases heat in the core, where it cools before being sent through the process once more.

Definition 1.2.6. Seals important engine components with the timing head/cover *gasket*. prevents the blending of cylinder pressure, antifreeze, and oil.

Definition 1.2.7. The $radiator\ overflow\ tank$ is a plastic container with one overflow hole and one intake that is often installed adjacent to the radiator. The identical tank that you fill with water before driving is this one.

Definition 1.2.8. The cooling system can be kept pressured by the *pressure cap's* assistance in sealing the system. To keep the coolant in the radiator from boiling, it is pressured. The system remains more effective as a result.

Definition 1.2.9. A network of rubber *hoses* transports coolant from the engine to the radiator. Furthermore, after years of usage, these hoses may begin to leak.

1.3 Automobiles Coolant

Definition 1.3.1. *Car coolant* is a liquid combination often made up of water and particular chemical additives that is used to control and maintain an internal combustion engine's operating temperature. It moves throughout the cooling system and engine to disperse extra heat produced during combustion and stop the engine from overheating or freezing in cold weather.

Definition 1.3.2. *Ethylene glycol*, usually known as antifreeze, is frequently utilized in automobile coolants. It is a crucial ingredient in the creation of coolant/antifreeze since it aids in controlling the engine's temperature and guards against freezing in the winter and overheating in the summer.

Definition 1.3.3. *Propylene glycol*, also known as *PG* is a synthetic organic chemical that is frequently used as an antifreeze or coolant in a variety of applications, such as automotive engines, heating and cooling systems, and food processing. It belongs to the glycol class of organic compounds, which is recognized for its capacity to both lower the freezing point and raise the boiling point of water.

Definition 1.3.4. *Diethylene glycol* sometimes known as *DEG* is a chemical that is used in several antifreeze and coolant compositions. This particular form of glycol, which is comparable to ethylene glycol and propylene glycol, has a number of characteristics that make it appropriate for use in cooling systems.

Definition 1.3.5. *Silicate* acts as a corrosion inhibitor in the cooling system. It forms a protective film or layer on the metal surfaces inside the engine, including the radiator, heater core, and engine block. This protective layer helps prevent the metal from corroding or rusting over time.

Definition 1.3.6. A *corrosion inhibitor* is a chemical or substance introduced to a system or environment to stop or slow down corrosion, which is the deterioration or degradation of metals and other materials exposed to the environment. Rust and other kinds of corrosion are created when metal surfaces react with ambient elements such oxygen, moisture, chemicals, and pollutants.

1.4 Laplace Transform

Pierre Simon De Laplace, who invented the Laplace transform, is the reason why it has that name.

Definition 1.4.1. $Laplace\ transformation$, like many other transformations, uses a set of principles to change one form of signal into another. When referring to a function, the Laplace transform is written as Lf(t) or F(s). The Laplace transform converts an equation into its algebraic form, which helps in solving problems. It operates by transforming a time-dependent (t) function into one or more complex-valued functions.The laplce transform of function $f(t)$ for $t \geq 0$ is defined as,

$$F(s) = \int_0^{+\infty} f(t)e^{-st}dt,$$

where s is either real or complex parameter.

Definition 1.4.2. Applying the $inverse\ laplace\ transform$ to $F(s)$ will return the function to its original state. Laplace $F(s)$'s inverse is as follows.

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

For instance, if we know its Laplace transform, we may find the original function.

Application of Laplace Transform

The Laplace transform is used in the examples below.

- The Laplace transformation offers the benefit of obtaining a solution for an ordinary differential equation, complete with appropriate initial conditions, without the need to first determine the general form and compute arbitrary constants based on those initial conditions.

- In the realm of broadcast communications, this technique serves to send signals to both ends of the communication channel. To illustrate, when signals are sent via a telephone system, they undergo a conversion process into a oscillating wave pattern prior to their transmission over the medium.

- We were able to convert the ODE solution into an algebraic equation using this technique.

1.5 Hankel Transform

Any function $f(r)$ in is represented by the $Hankel\ transform$ as the weighted sum of an essentially infinite number of $J_v(kr)$-type Bessel functions. The six Bessel functions that make up the total all have the same order, ν, but differ in the r axis by a scaling factor, k. The modified function was formulated by incorporating the necessary coefficient, denoted as F_v, into the summation, and this was expressed in relation to the scaling factor, denoted as k. Hermann Hankel is credited with the development of the Hankel transform, a type of integral transformation that is also commonly referred to as the Fourier-Bessel transform.

Definition 1.5.1. Consider the function $q(r)$ defined for $r \geq 0$. The Hankel transformation of $q(r)$ with a specified order, can be expressed as follows:

$$Q(r) = H_w\{q(r)\} = \int_0^{+\infty} rq(r)J_w(kr)dr.$$

where J_w is first kind of Bessel function.

Definition 1.5.2. The *inverse hankel transform* of $Q_w(r)$ is expressed as,

$$q(r) = \int_0^{+\infty} kQ_w(k)J_w(kr)dr.$$

1.6 Non-Dimensionalization

Definition 1.6.1. The technique of fully or partially eliminating dimensions by substituting suitable variables for the physical values included in the equation. When approximated units are involved, this technique can reorganize and characterize the problems. It is clearly associated with dimensional analysis.

Steps for non dimensionalization

- Identify the dependent and independent variables
- Replace with a scaled quantity for a certain unit of measure.
- The largest order polynomial or derivative term should be divided by its coefficient.
- Choose each factor's characteristic unit specification carefully so that its coefficient becomes 1.
- Once again, construct the system of equations using non-dimensional quantities.

Non-dimensionalization reduces the number of tests required to test a hypothesis, helps with working on differential equations, rescales components to a unit-less structure, eliminates redundant borders, and more.

Definition 1.6.2. The *Prandtl number* (Pr) is a dimensionless parameter and a fundamental characteristic of fluids. The ratio is between momentum diffusivity ν and heat diffusivity α. Small Prandtl numbers are free-flowing fluids with high thermal conductivity values, making them a good choice for heat-conducting fluids. The Prandtl number is represented mathematically as

$$Pr = \frac{\nu}{\alpha}$$

Definition 1.6.3. In fluid dynamics, the *Reynolds number* (Re), a dimensionless quantity, is used to describe the flow regime of a fluid in a pipe, channel, or across a surface.We have specific Reynolds number values that indicate the kind of flow, such as $Re < 2000$ indicated laminar flow, there is turbulent flow when $Re > 4000$ and transition range of Re is for values between 2000 and 4000.Although the flow of coolant through the radiator has a Reynolds number of 20,000. This value falls within the range of turbulent flow (Re ¿ 4,000), indicating that the radiator's internal flow is turbulent. Because turbulent flow is better for effective engine cooling, the Reynolds number for automotive coolant flow through the radiator is utilized to determine this.

Definition 1.6.4. In fluid mechanics and heat transfer, the Grashof number (Gr) is a dimensionless number that is used to describe the effects of buoyancy in a fluid flow or heat transfer process, particularly in examples involving natural convection.

$$G_r = \frac{g\beta(T_\omega - T_\infty)R^3}{\nu^2}$$

where gravitational acceleration denoted by g, β represent the thermal expansion coefficient, T_ω the surface's temperature, T_∞ temperature of the fluid far from the surface, ν kinetic viscosity.

Definition 1.6.5. The *Solute Grashof number* (G_m) , a dimensionless number used in fluid dynamics and heat and mass transfer, quantifies the relative importance of buoyancy-driven flow to molecular diffusion of a solute (material being transported) within a fluid. It is specifically applied to circumstances where fluid motion is caused by density fluctuations resulting from variations in solute concentration.

Definition 1.6.6. The *Peclet number* (Pe), a dimensionless number, is employed to assess the relative importance of advection and diffusion in fluid mechanics and heat transfer. Diffusion is the spreading or mixing of a property as a result of concentration differences, whereas advection is the transport of a property by the bulk motion of the fluid.The French physicist Jean Claude Eugène Péclet is honored with its name.

$$Pe = \frac{UL}{\alpha}$$

where U is the flow of velocity, length denoted by L, α thermal diffusion.

Additionally, in the context of thermal energy exchange.,

$$Pe = Pr \cdot Re$$

Definition 1.6.7. The *Soret number* (Sr) measures the ratio of temperature difference to concentration. Soret's number has a high value since it represents a wide temperature range and steep gradient.

Definition 1.6.8. The *Schmidt number* (Sc), a dimensionless number, is used in fluid mechanics and heat and mass transfer to describe the relative importance of molecular diffusion and momentum diffusion in a fluid. Molecular diffusion is the spreading or mixing of molecules as a result of concentration differences.

Definition 1.6.9. The *Darcy number* (Da),proposed by Henry Darcy is defines as the ratio between the materials permeability (K) to square of its cross-sectional region (d). As

$$Da = \frac{K}{d^2}$$

where K stands for the medium's permeability and d for the length.

Definition 1.6.10. A dimensionless quantity the *Hartmann number* (Ha) introduced by Julius Hartmann defined as the square root of the ratio of electromagnetic force to viscous force. When researching how fluids move in magnetic fields, it is frequently dealt with. Mathematically,

$$Ha = \sqrt{\frac{Electromagnetic\ force}{viscous\ force}}$$

This often happens when fluids are moving across magnetic fields. Here is the Hartmann number:

$$Ha = BL\sqrt{\frac{\sigma}{\mu}}$$

where B denotes the strength of the magnetic field, L denotes the characteristic length scale,σ denotes electrical conductivity, and μ denotes dynamic viscosity figuratively.

Chapter 2

Introduction

In the evolution of the modern automobile, the German engineer Karl Benz is largely credited with creating the first gasoline-powered tricycle in 1886. The vehicle has a water cooling system and a two-stroke gasoline engine, as well as several standard features found in contemporary vehicles. However, there is no radiator; instead, a long, curved conduit circulates water to keep the internal combustion engine cool. This form of coiled radiator is far from meeting demand because it is inefficient, uses a lot of water, and necessitates a large cooling circuit.The Daimler Engine Company was established by Daimler and Maybach in 1890. The Phoenix engine, which Maybach, Daimler, and Daimler's son created in 1894, set the stage for the adoption of four-cylinder engines in automobiles. Maybach is simultaneously making great efforts to address the issue of engine heat dissipation and has finished several inventions including tubular radiators with fans and honeycomb radiators.Maybach made a crucial advancement toward resolving the engine cooling issue in 1897. He suggested a honeycomb construction made up of a lot of little square tubes. Copper wires were put in the space between the tubes to provide a conduit for coolant flow. Together, solder. To cool the coolant, air is chilled inside the tube. This is an effective heat exchanger that can cut back on coolant usage dramatically. Maybach created the new Mercedes 35HP racing automobile in 1900. At the year's end, the prototype was ready. It features a 35 horsepower engine with a top speed of 75 kph. The next year, the vehicle was displayed at the second auto show in the United States and generated a lot of buzz. Jelinek used it to win numerous world championships throughout 1901. This automobile's official debut as a manufacturing model occurred in 1902. The car was coveted by Europe's upper class as soon as it hit the market because to its unbeatable performance on the field, and it garnered a significant amount of orders. The first modern car in history was the Mercedes 35HP, according to some. It comprises a variety of inventive characteristics, such as a front-mounted engine and radiator, a sleek design, a low center of mass, an extended wheelbase, a tilted steering column, equally-sized wheels at both the front and rear, and numerous additional elements.This vehicle is widely regarded as the inaugural modern automobile, marking the conclusion of the "motorized carriage" era. Consequently, Maybach is often hailed as the pioneer of automotive design. To address heat dissipation for the 5.9-liter inline four-cylinder gasoline engine, the Mercedes 35HP incorporated a honeycomb radiator, an innovation credited to Maybach, at the front of the vehicle[1].

The radiator is made up of up to 8,070 tiny square tubes, each measuring $66mm$ in size. A rectangular radiator core is created by soldering together. A little fan is installed behind the radiator to enhance cooling performance at low flow rates. Inefficient coil heat sinks were once employed in automobiles.Later, he converted the square tube to a round tube to improve the design. A hexagon was formed by piercing the tube's two ends, and the ends were then soldered together to form a solid unit. Keep your copper wire. However, the honeycomb radiator's internal water flow speed is slow, making it simple for water to accumulate inside of it. Additionally, the welding seam is long, making it simple for leaks to form. Consequently, a tube-fin radiator was later installed in its stead.

The most frequently discussed subjects in the industries are fluid flow and heat transfer. Fluid heat transfer is used by a variety of equipment, including heat exchangers, solar collectors, boilers, and similar systems [2-7]. Antifreeze is a fluid that is crucial for the proper operation of solar collectors, outdoor heat exchangers, and automobiles during the colder months of the year. Ethylene glycol (EG) and water can be mixed to produce a solution known as antifreeze. This solution effectively reduces the freezing point and elevates the boiling point when employed as the working fluid in the mentioned system. Normally, a 50% water and 50% EG combination is utilized to safeguard against frost, which can lower the freezing point to $-37°C$. Furthermore, this combination raises the boiling point to 107°C. Although this mixture is well-regarded for its ability to protect equipment during the winter, it possesses limited thermal conductivity. As an example, at a temperature of 25°C, pure water exhibits a thermal conductivity of approximately 0.60, whereas a mixture demonstrates a thermal conductivity of about 0.37. Recent research efforts have focused on augmenting the heat transfer capabilities of antifreeze by transforming it into a nanofluid [8–10]. Multiple investigations have indicated that nanofluids exhibit superior thermal conductivity when compared to conventional fluids [11–17].

Typical conductive fluids encompass coolants like ethylene glycol and water, lubricants such as oils, paraffin, biofluids, and polymer solutions. Nanofluids, when contrasted with conventional fluids, exhibit superior performance in attributes like viscosity, heat transfer rate, thermal diffusion, and thermal conductivity. Nanofluids are used in a wide range of healthcare, technical, and industrial applications due to their increased properties including,smart fluids,magnetic drugs, electronics cooling,nanocryosurgery, vehicle cooling,heat transfer and industrial cooling [18]. The most modern uses of heat transmission include pipelines, electric conductors, solar collectors, and heat exchangers. These applications, which rely on convection, are used to lower and increase system heat. It is evident that the proper fluids are necessary for heat transfer. Nevertheless, it is clear that the right fluids are crucial for heat transmission.

As was already mentioned, nanoparticles can increase a fluid's thermal conductivity, but it is important to keep in mind that their presence may also alter other thermophysical characteristics of the fluid, which isn't always a good thing. Viscosity plays a crucial role when examining the heat transfer of fluids among these attributes. Viscosity is a factor in the majority of the variables used to model how heat moves through fluids. Therefore, researchers have been interested in measuring the nanofluid's viscosity.

There is a limited body of research concerning non-Newtonian characteristics, even though extensive studies have been conducted regarding the viscosity of nanofluids. For example, in the work of Eshgarf and Afrand [19], an exploration into the viscosity of antifreeze in the presence of hybrid nanomaterials composed of silica and carbon nanotubes was undertaken. Nano-antifreeze samples were created with solid volume fractions ranging from 0.0625% to 2%. These experiments encompassed a range of shear rates and were conducted at room temperature, extending up to 50°C. The results revealed that while all nano-antifreeze samples exhibited non-Newtonian fluid behavior, pure antifreeze displayed entirely Newtonian behavior. Notably, the viscosity decreased as the shear rate increased, indicating a shear-thinning behavior in the nanofluid. Additionally, a consistency score and a power law index were determined by examining the relationship between shear stress and shear rate.The researchers observed that as the solid volume percentage increased, the consistency index increased while the power law index decreased. To predict the consistency index and power law index, they eventually proposed two correlations. In a similar vein, Afrand and colleagues[20] delved into the rheological behavior of ethylene glycol containing hybrid nanoparticles of iron oxide and silver by testing viscosity at various shear rates. The study also investigated how temperature and nanoparticle concentration influenced viscosity. Their research revealed that nanofluids behaved as Newtonian fluids when their volume fractions were below 0.3%, but they exhibited non-Newtonian behavior when the volume fractions exceeded 0.3%. It was observed that more concentrated nanofluids adhered to a power law model, and to determine the model's parameters, a curve fitting technique was employed. Shahs Avani et al.[21] conducted a study on the rheological properties of antifreeze containing functionalized multi-walled carbon nanotubes. They examined the viscosity of various nano-antifreeze samples under different shear rates and temperatures. Their findings revealed that the antifreeze behaved as a Newtonian fluid, while the other nano-antifreeze samples exhibited non-Newtonian behavior following a power law model with fluctuations in viscosity at varying shear rates[22]. The researchers determined the consistency index and power law index through curve fitting and noted that the consistency index increased with higher volume fractions and decreased temperatures. They eventually offered empirical connections to forecast the power law model constants. The ability of silicate compounds to absorb heat in a car radiator is typically not quantified in the same way that a substance's specific heat capacity may be. Instead, the radiator's capacity for heat absorption and dissipation is mostly determined by its structure and components, and its relevance in the cooling system is primarily associated with corrosion prevention [23].

The engine in your car performs best when the temperature is high. When the engine is cold, emissions rise, efficiency declines, and component deterioration occurs more quickly. Therefore, the cooling system's major goals are to speed up engine warm-up and maintain a stable operating temperature. The cooling system's primary responsibility is to maintain the engine at the proper temperature. If any component of the cooling system malfunctions, the engine could get too hot and suffer serious repercussions. Have you ever thought about what might happen if your cooling system breaks down? Overheating can lead to serious problems including cracked engine blocks and failed cylinder head gaskets. Heat can also fuse the pistons inside the cylinders, requiring the replacement of the engine as a whole. Therefore, it is essential to know how your engine's cooling system works and to provide it regular maintenance [24]. It's crucial to start by defining a cooling system's objective in order to fully understand how it works. The main function of an automobile's cooling system is to control the engine's temperature and guard against overheating. This process may seem fairly difficult given the significant heat produced by an automotive engine. In order to put it into perspective, when a car is moving at 50 mph on a highway, its engine produces about 4,000 combustion events each minute. This ongoing process generates a significant quantity of heat, which is exacerbated by the friction caused by moving engine parts. The engine would quickly overheat and shut down without a trustworthy cooling system. A contemporary cooling system is built to keep the engine's temperature within the proper range, regardless of whether it is 115 degrees outside or it is the dead of winter [25]. Cooling system operates by continuously circulating coolant through channels within the engine block. A water pump drives the movement of coolant through the cylinder block, where it actively absorbs heat from the engine.Subsequently, this heated fluid exits the engine and travels to the radiator, where it is cooled by the airflow from the vehicle's radiator grill. Upon cooling, the fluid is directed back to the engine as it passes through the radiator once more, effectively carrying away additional engine heat.Between the radiator and the engine, a thermostat is positioned to regulate the response of the coolant to temperature changes. When the fluid temperature drops below a predetermined level, it bypasses the radiator and returns to the engine block. However, when the coolant reaches a specific temperature, the thermostat's valve opens, allowing it to flow through the radiator for further cooling.Due to the high operating temperature of the engine, the coolant can potentially reach its boiling point rapidly. To prevent this, the system is pressurized. Increased pressure makes it more challenging for the coolant to boil. Occasionally, pressure may build up in the hose or gasket, necessitating release to avoid damage. The radiator cap collects any excess fluid and pressure and directs it into a reserve tank. Once the liquid in the storage tank has cooled to an appropriate temperature, it is reintroduced into the cooling system [26].

Every automobile coolant contains glycol.Glycol comes in a variety of forms, but we only used three in coolant formation: ethylene glycol, propylene glycol, and diethylene glycol.Due to its higher thermal properties, ethylene glycol is frequently regarded as a more effective coolant than propylene glycol and diethylene glycol in some applications [27].Comparing ethylene glycol to propylene and diethylene glycol, ethanol has a larger heat capacity and thermal conductivity. Because of its improved ability to absorb and transport heat, it is a better heat transfer fluid in many cooling systems. Propylene and diethylene glycol have higher freezing points than ethylene glycol, which makes ethylene glycol a better choice for uses in colder climes. It offers superior defense against freezing and possible cooling system harm.Additionally, ethylene glycol has a greater boiling point than diethylene and propylene glycol. This makes it suited for high-temperature applications since it enables it to work at higher temperatures without boiling and losing its cooling efficiency. Because of its greater heat transmission capabilities, ethylene glycol is frequently preferred in automotive engines where maintaining a stable operating temperature is essential. It enhances engine efficiency and aids in the prevention of overheating.Propylene glycol and diethylene glycol are often more expensive than ethylene glycol, which makes it a sensible option for many industrial uses[28]. Common coolants are made up of ethylene glycol, water, and additive packages. Propylene glycol plus water make up another glycol-based coolant. Propylene glycol is less poisonous than the other variety, which is the main distinction between them. If the only factor in choosing a coolant was its capacity to dissipate heat, pure water would be the greatest choice because it has a higher heat-carrying capacity than pure ethylene glycol. But there are other difficulties with water. Rust develops on iron engine components. The rust is subsequently transported to other cooling regions. Even before the accumulation clogs the radiator and overflows the cooling system with dirt, the ensuing corrosion obstructs the transfer of heat. Coolant serves to reduce engine rust and corrosion while also providing antifreeze capabilities. Unlike water, it does not expand and freeze in extremely cold weather, thus preventing pressure buildup and engine cracking.

Chapter 3

Mathematical modeling of problem

3.1 Model of the Problem

Suppose that coolant is flowing in an unstable manner inside of a cylindrical round tube, with the radial axis being r and the cylindrical axis being z. It is considered that ethylene glycol behaves as an incompressible non-newtonian type of fluid but non-newtonian fluid through the capillaries tubes behave as newtonian fluid.
The momentum equation can be expressed as follows under the conditions stated above, such as [29]-[30]

$$\frac{\partial \bar{u}}{\partial \bar{t}} = -\frac{-1}{\rho}\frac{\partial \bar{p}}{\partial \bar{z}} + \nu(\frac{\partial^2 \bar{u}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{u}}{\partial \bar{r}}) + G(\bar{t}) + g\beta_T(\bar{T} - \bar{T_\infty}) + g\beta_C(\bar{C} - \bar{C_\infty}) \tag{3.1.1}$$

where $\bar{T}$ is the coolant's temperature and $\bar{u}$ stands for the axial velocity.$\bar{C}$ stands for the concentration of coolant.β_T and β_C are the corresponding coefficients of thermal expansion and concentration expansion, where ρ represents density and g is the acceleration due to gravity.
The Navier-Stokes equation does not include the convective term $u(\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y})$ in the model under consideration due to its modest Reynolds number stream of macro scale length. However, the convective term cannot be discounted in micro-channel flow. Assuming that the transverse velocity is lower than the axial velocity, we assumed that unidirectional coolant would flow in an axial direction. the body's rate of ac-

celeration in a vibrating environment, such as in The body of car's acceleration in a vibratory environment like [31],[32] [33], can take the following forms:

$$G(\bar{t}) = \bar{A_0}(\cos(\bar{k}\bar{t} - \phi_0)$$

where $\bar{A_0}$ is the body's amplitude of acceleration, $\bar{k}$ denotes frequency, and ϕ_0 denotes phase angle.

Now,

$$\frac{\partial \bar{u}}{\partial \bar{t}} = -\frac{-1}{\rho}\frac{\partial \bar{p}}{\partial \bar{z}} + \nu(\frac{\partial^2 \bar{u}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{u}}{\partial \bar{r}}) + \bar{A_0}(\cos(\bar{k}\bar{t} - \phi_0) + g\beta_T(\bar{T} - \bar{T_\infty}) + g\beta_C(\bar{C} - \bar{C_\infty}) \tag{3.1.2}$$

The energy equation in the presence of thermal radiation, used in [30],[34] can be written as

$$\frac{\partial \bar{T}}{\partial \bar{t}} = \frac{K}{\rho c_p}(\frac{\partial^2 \bar{T}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{T}}{\partial \bar{r}}) - \frac{1}{\rho c_p}\frac{\partial \bar{q}}{\partial \bar{r}} + \frac{\bar{Q_m} + \bar{\theta}_m}{\rho c_p} \tag{3.1.3}$$

where c_p is specific heat and K is thermal conductivity. Heat is produced by metabolism $\bar{Q_m}$, and heat is absorbed $\bar{\theta_m}$. so that the heat flux is about(cf.[30])

$$-\frac{\partial \bar{q_r}}{\partial \bar{r}} = 4\beta_1^2(\bar{T} - \bar{T_\infty})$$

on the above equation $\beta_1^2{=}\int_0^\infty \frac{\partial B}{\partial \bar{T}}$, here ξ represents the factor related to radiation absorption, Ω denotes the frequency where as B stands for Planck's constant.

we get,

$$\frac{\partial \bar{T}}{\partial \bar{t}} = \frac{K}{\rho c_p}(\frac{\partial^2 \bar{T}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{T}}{\partial \bar{r}}) - \frac{1}{\rho c_p}4\beta_1^2(\bar{T} - \bar{T_\infty}) + \frac{\bar{Q_m} + \bar{\theta}_m}{\rho c_p} \tag{3.1.4}$$

The concentration equation with respect to the Soret effect can be noted,such as [35]

$$\frac{\partial \bar{C}}{\partial \bar{t}} = D_m(\frac{\partial^2 \bar{C}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{C}}{\partial \bar{r}}) + \frac{D_m K_T}{T_\infty}(\frac{\partial^2 \bar{T}}{\partial^2 \bar{r}} + \frac{1}{\bar{r}}\frac{\partial \bar{T}}{\partial \bar{r}}) \tag{3.1.5}$$

Where D_m indicates the mass's diffusibility , K_T represents a proportion of thermal diffusion. , and T_∞ denotes the surrounding temperature.

The boundary conditions for this particular model are defined as follows,

$$\begin{aligned} &\bar{C}(\bar{r}, 0) = 0, \bar{T}(\bar{r}, 0) =, \bar{u}(\bar{r}, 0) = 0 \qquad at \;\; \bar{t} = 0, \bar{r} \in [0, R_0] \\ &\frac{\partial \bar{T}}{\partial \bar{r}} = 0, \frac{\partial \bar{C}}{\partial \bar{r}} = 0, \frac{\partial \bar{u}}{\partial \bar{r}} = 0 \qquad at \;\; \bar{r} = 0 \\ &\bar{T}(R_0, \bar{t}) = T_\omega, \bar{u}(R_0, \bar{t}) = 0, \bar{C}(R_0, \bar{t}) = C_\omega \end{aligned} \tag{3.1.6}$$

Now, the following dimensionless parameters are introduced as,

$$
\begin{aligned}
&r=\frac{\bar{r}}{R_0}, u=\frac{\bar{u}}{u_0}, z=\frac{\bar{z}}{R_0}, p=\frac{\bar{p}}{u_0^2\rho}, t=\frac{\bar{t}}{R_0u_0}, A_0=\frac{\bar{A_0}}{R_0u_0^2}\\
&k=\frac{\bar{k}}{R_0u_0}, \theta=\frac{\bar{T}-T_\infty}{T_\omega-T_\infty}, \phi=\frac{\bar{C}-C_\infty}{C_\omega-C_\infty}, Q_m=\frac{R_0\bar{Q_m}}{u_0\rho c_p(T_\omega-T_\infty)},\\
&\theta_m=\frac{R_0\bar{\theta_m}}{u_0\rho c_p(T_\omega-T_\infty)}
\end{aligned}
\tag{3.1.7}
$$

Utilizing the non-dimensional variables described earlier then equation (3.1.1) will become,

$$
\begin{aligned}
\frac{u_0\partial u}{\frac{R_0}{u_0}\partial t}=&-\frac{-1}{\rho}\frac{\rho u_0^2\partial p}{R_0\partial z}+\nu(\frac{u_0\partial^2 u}{R_0^2\partial^2 r}+\frac{1}{R_0 r}\frac{u_0\partial u}{R_0\partial r})+\frac{A_0u_0^2}{R_0}(\cos(\frac{ku_0}{R_0}\frac{tR_0}{u_0}-\phi_0)\\
&+g\beta_T[\theta(T_\omega-T_\infty)+T_\infty-T_\infty]+g\beta_C[\phi(C_\omega-C_\infty)+C_\infty-C_\infty]
\end{aligned}
$$

$$
\begin{aligned}
\frac{u_0^2\partial u}{R_0\partial t}=&-\frac{u_0^2\partial p}{R_0\partial z}+\nu\frac{u_0}{R_0^2}(\frac{\partial^2 u}{\partial^2 r}+\frac{1}{r}\frac{\partial u}{\partial r})+\frac{A_0u_0^2}{R_0}(\cos(kt-\phi_0)\\
&+g\beta_T[\theta(T_\omega-T_\infty)]+g\beta_C[\phi(C_\omega-C_\infty)]
\end{aligned}
$$

$$
\begin{aligned}
\frac{u_0^2\partial u}{R_0\partial t}=&\frac{u_0^2}{R_0}(-\frac{\partial p}{\partial z}+\frac{\nu}{u_0R_0}(\frac{\partial^2 u}{\partial^2 r}+\frac{1}{r}\frac{\partial u}{\partial r})+A_0(\cos(kt-\phi_0)\\
&+\frac{g\beta_T[\theta(T_\omega-T_\infty)]R_0}{u_0^2}+\frac{g\beta_C[\phi(C_\omega-C_\infty)]R_0}{u_0^2})
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial u}{\partial t}=&-\frac{\partial p}{\partial z}+\frac{\nu}{u_0R_0}(\frac{\partial^2 u}{\partial^2 r}+\frac{1}{r}\frac{\partial u}{\partial r})+A_0(\cos(kt-\phi_0)\\
&+\frac{R_0^2\nu^2}{R_0^2\nu^2}\frac{g\beta_T[\theta(T_\omega-T_\infty)]R_0}{u_0^2}+\frac{R_0^2\nu^2}{R_0^2\nu^2}\frac{g\beta_C[\phi(C_\omega-C_\infty)]R_0}{u_0^2}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial u}{\partial t}=&-\frac{\partial p}{\partial z}+\frac{\nu}{u_0R_0}(\frac{\partial^2 u}{\partial^2 r}+\frac{1}{r}\frac{\partial u}{\partial r})+A_0(\cos(kt-\phi_0)\\
&+\frac{\nu^2}{R_0^2u_0^2}\frac{g\beta_T[\theta(T_\omega-T_\infty)]R_0^3}{\nu^2}+\frac{\nu^2}{R_0^2u_0^2}\frac{g\beta_C[\phi(C_\omega-C_\infty)]R_0^3}{\nu^2}
\end{aligned}
$$

$$
\frac{\partial u}{\partial t}=-\frac{\partial p}{\partial z}+\frac{1}{Re}(\frac{\partial^2 u}{\partial^2 r}+\frac{1}{r}\frac{\partial u}{\partial r})+A_0(\cos(kt-\phi_0)+\frac{G_r\theta}{Re^2}+\frac{G_m\phi}{Re^2}
$$

The movement characteristic of the coolant begins with the characteristic of the waterpump, hence the term for pressure gradient is a particular type of,

$$-\frac{\partial p}{\partial z} = b_0 + b_1 \cos(\omega t)$$

Then we get,

$$\frac{\partial u}{\partial t} = b_0 + b_1 \cos(\omega t) + \frac{1}{Re}(\frac{\partial^2 u}{\partial^2 r} + \frac{1}{r}\frac{\partial u}{\partial r}) + A_0(\cos(kt - \phi_0) + \frac{G_r \theta}{Re^2} + \frac{G_m \phi}{Re^2} \quad (3.1.8)$$

Dimensionless quantities are substituted in the equation (3.1.4) we get,

$$\frac{u_0(T_\omega - T_\infty)}{R_0}\frac{\partial \theta}{\partial t} = \frac{K}{\rho c_p}\left(\frac{(T_\omega - T_\infty)}{R_0^2}\frac{\partial^2 \theta}{\partial^2 r} + \frac{(T_\omega - T_\infty)}{R_0 r}\frac{\partial \theta}{\partial \bar{r}}\right) + \frac{1}{\rho c_p}4\beta_1^2(T\omega - T_\infty)\theta + \frac{u_0 \rho c_p(Q_m + \theta_m)((T_\omega - T_\infty))}{\rho c_p}$$

$$\frac{u_0(T_\omega - T_\infty)}{R_0}\frac{\partial \theta}{\partial t} = \frac{K(T_\omega - T_\infty)}{\rho c_p R_0^2}[(\frac{\partial^2 \theta}{\partial^2 r} + \frac{1}{r}\frac{\partial \theta}{\partial r}) + \frac{R_0^2 4\beta_1^2 \theta}{K} + \frac{u_0 R_0 \rho c_p(Q_m + \theta_m)}{K}]$$

$$u_0\frac{\partial \theta}{\partial t} = \frac{K}{\rho c_p R_0}\left[(\frac{\partial^2 \theta}{\partial^2 r} + \frac{1}{r}\frac{\partial \theta}{\partial r}) + \frac{R_0^2 4\beta_1^2 \theta}{K} + \frac{u_0 R_0 \rho c_p(Q_m + \theta_m)}{K}\right]$$

$$\frac{u_0 \rho c_p R_0}{K}\frac{\partial \theta}{\partial t} = \left(\frac{\partial^2 \theta}{\partial^2 r} + \frac{1}{r}\frac{\partial \theta}{\partial r}\right) + \frac{R_0^2 4\beta_1^2 \theta}{K} + \frac{u_0 R_0 \rho c_p(Q_m + \theta_m)}{K}$$

$$\frac{u_0 \mu c_p R_0}{K\nu}\frac{\partial \theta}{\partial t} = \left(\frac{\partial^2 \theta}{\partial^2 r} + \frac{1}{r}\frac{\partial \theta}{\partial r}\right) + \frac{R_0^2 4\beta_1^2 \theta}{K} + \frac{u_0 R_0 \mu c_p(Q_m + \theta_m)}{K\nu}$$

$$RePr\frac{\partial \theta}{\partial t} = \left(\frac{\partial^2 \theta}{\partial^2 r} + \frac{1}{r}\frac{\partial \theta}{\partial r}\right) + R\theta + RePr(Q_m + \theta_m) \quad (3.1.9)$$

By replacing dimensionless parameters in equation (3.1.5) we obtain:

$$\frac{u_0(C_\omega - C_\infty)}{R_0}\frac{\partial \phi}{\partial t} = D_m\left[\frac{(C_\omega - C_\infty)}{R_0^2}\frac{\partial^2 \phi}{\partial r^2} + \frac{(C_\omega - C_\infty)}{R_0^2 r}\frac{\partial \phi}{\partial r}\right] + \frac{D_m K_t}{T_\infty}\left[\frac{(T_\omega - T_\infty)}{R_0^2}\frac{\partial^2 \theta}{\partial r^2} + \frac{(T_\omega - T_\infty)}{R_0^2}\frac{\partial \theta}{\partial r}\right]$$

$$\frac{u_0(C_\omega - C_\infty)}{R_0}\frac{\partial\phi}{\partial t} = \frac{D_m(C_\omega - C_\infty)}{R_0^2}[(\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}) + \frac{K_t(T_\omega - T_\infty)}{T_\infty(C_\omega - C_\infty)}(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r})]$$

$$\frac{u_0R_0}{D_m}\frac{\partial\phi}{\partial t} = \left(\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}\right) + \frac{K_t(T_\omega - T_\infty)}{T_\infty(C_\omega - C_\infty)}\left(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r}\right)$$

$$\frac{u_0R_0}{\nu}\frac{\nu}{D_m}\frac{\partial\phi}{\partial t} = \left(\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}\right) + \frac{D_mK_t(T_\omega - T_\infty)}{\nu T_\infty(C_\omega - C_\infty)}\frac{\nu}{D_m}\left(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r}\right)$$

$$ReSc\frac{\partial\phi}{\partial t} = \left(\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}\right) + SrSc\left(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r}\right) \tag{3.1.10}$$

After presenting and resolving the equation using dimensionless parameters,

$$\frac{\partial u}{\partial t} = b_0 + b_1cos(\omega t) + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial^2 r} + \frac{1}{r}\frac{\partial u}{\partial r}\right) + A_0(cos(kt - \phi_0) + \frac{G_r\theta}{Re^2} + \frac{G_m\phi}{Re^2}$$
$$RePr\frac{\partial\theta}{\partial t} = \left(\frac{\partial^2\theta}{\partial^2 r} + \frac{1}{r}\frac{\partial\theta}{\partial r}\right) - R\theta + RePr(Q_m + \theta_m)$$
$$ReSc\frac{\partial\phi}{\partial t} = \left(\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}\right) + SrSc\left(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r}\right)$$

where Thermal Grashof number Gr, Solutal Grashof number Gm,Reynolds number Re, Schmidt number Sc, Prandtl number Pr, Soret number Sr ,radiation parameter R and Peclet number Pe are defined as

$$Gr = \frac{g\beta_T[(T_\omega - T_\infty)]R_0^3}{\nu^2}, Gm = \frac{g\beta_C[(C_\omega - C_\infty)]R_0^3}{\nu^2}, Re = \frac{R_0u_0}{\nu}$$
$$Sc = \frac{\nu}{D_m}, Pr = \frac{\mu c_p}{K}, Sr = \frac{D_mK_t(T_\omega - T_\infty)}{\nu T_\infty(C_\omega - C_\infty)}, R = \frac{R_0^2 4\beta_1^2}{K}, Pe = Re{\cdot}Pr$$

These are the boundary conditions in dimensionless form,

$$\begin{aligned} &u(r,0) = 0,\ \theta(r,0) = 0,\ \phi(r,0) = 0, && at\ t = 0,\ r \in [0,1] \\ &\frac{\partial u}{\partial r} = 0,\ \frac{\partial\theta}{\partial t} = 0,\ \frac{\partial\phi}{\partial} = 0 && at\ r = 0 \\ &u(1,t) = 0,\ \theta(1,t) = 0,\ \phi(1,t) = 0 && at\ t > 0 \end{aligned} \tag{3.1.11}$$

3.2 Solution procedure

By applying the Laplace transform to the equations with respect to the time variable, considering the initial conditions, we can express from the equations (3.1.8)-(3.1.10).

$$\mathcal{L}\{\frac{\partial u}{\partial t}\} = \mathcal{L}\{b_0 + b_1\cos(\omega t) + \frac{1}{Re}(\frac{\partial^2 u}{\partial^2 r} + \frac{1}{r}\frac{\partial u}{\partial r}) + A_0(\cos(kt - \phi_0) + \frac{G_r\theta}{Re^2} + \frac{G_m\phi}{Re^2}\}$$

$$\begin{aligned} su(r,s) =& \frac{b_0}{s} + b_1\frac{s}{s^2+\omega^2} + \frac{1}{Re}\left(\frac{\partial^2 u(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial u(r,s)}{\partial r}\right) + A_0\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right) \\ &+ \frac{Gr\theta(r,s)}{Re^2} + \frac{Gm\phi(r,s)}{Re^2} \end{aligned}$$

$$u(r,s) = \frac{b_0}{s^2} + \frac{b_1}{s^2+\omega^2} + \frac{1}{sRe}\left(\frac{\partial^2 u(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial u(r,s)}{\partial r}\right) + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right) \tag{3.2.1}$$

$$+ \frac{Gr\theta(r,s)}{sRe^2} + \frac{Gm\phi(r,s)}{sRe^2}$$

$$\mathcal{L}\{Pe\frac{\partial \theta}{\partial t}\} = \mathcal{L}\left((\frac{\partial^2\theta}{\partial^2 r} + \frac{1}{r}\frac{\partial\theta}{\partial r}) - R\theta + RePr(Q_m + \theta_m)\right)$$

$$sPe\bar{\theta}(r,s) = \left(\frac{\partial^2\bar{\theta}(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial\bar{\theta}(r,s)}{\partial r}\right) + R\bar{\theta}(r,s) + \frac{Pe(Q_m+\theta_m)}{s}$$

$$sPe\bar{\theta}(r,s) - R\bar{\theta}(r,s) = \left(\frac{\partial^2\bar{\theta}(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial\bar{\theta}(r,s)}{\partial r}\right) + \frac{Pe(Q_m+\theta_m)}{s}$$

$$(sPe - R)\bar{\theta}(r,s) = \left(\frac{\partial^2\bar{\theta}(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial\bar{\theta}(r,s)}{\partial r}\right) + \frac{Pe(Q_m+\theta_m)}{s} \tag{3.2.2}$$

$$\mathcal{L}\{ReSc\frac{\partial\phi}{\partial t}\} = \mathcal{L}\left((\frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r}) + SrSc(\frac{\partial^2\theta}{\partial r^2} + \frac{1}{r}\frac{\partial\theta}{\partial r})\right)$$

$$sReSc\bar{\phi}(r,s) = \left(\frac{\partial^2\bar{\phi}(r,s)}{\partial r^2} + \frac{1}{r}\frac{\partial\bar{\phi}(r,s)}{\partial r}\right) + \left(\frac{\partial^2\bar{\theta}(r,s)}{\partial^2 r} + \frac{1}{r}\frac{\partial\bar{\theta}(r,s)}{\partial r}\right) \tag{3.2.3}$$

Implementing the Hankel transformation to the aforementioned equations (3.2.1)-(3.2.3)

$$u_H(r_n,s) = \left[\frac{b_0}{s^2} + \frac{b_1 s}{s^2+\omega^2} + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} - \frac{r_n^2 u_H(r_n,s)}{sRe} + \frac{Gr\theta_H(r,s)}{sRe^2} + \frac{Gm\phi_H(r,s)}{sRe^2}$$

$$u_H(r_n,s) + \frac{r_n^2 u_H(r_n,s)}{sRe} = \left[\frac{b_0}{s^2} + \frac{b_1 s}{s^2+\omega^2} + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{sRe^2} + \frac{Gm\phi_H(r,s)}{sRe^2}$$

$$\left(1 + \frac{r_n^2}{sRe}\right) u_H(r_n,s) = \left[\frac{b_0}{s^2} + \frac{b_1 s}{s^2+\omega^2} + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{sRe^2} + \frac{Gm\phi_H(r,s)}{sRe^2}$$

$$\left(\frac{sRe + r_n^2}{sRe}\right) u_H(r_n,s) = \left[\frac{b_0}{s^2} + \frac{b_1 s}{s^2+\omega^2} + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{sRe^2} + \frac{Gm\phi_H(r,s)}{sRe^2}$$

$$u_H(r_n,s) = \left(\frac{sRe}{sRe + r_n^2}\right)\left[\frac{b_0}{s^2} + \frac{b_1 s}{s^2+\omega^2} + \frac{A_0}{s}\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{sRe^2} + \frac{Gm\phi_H(r,s)}{sRe^2}$$

$$u_H(r_n,s) = \left(\frac{Re}{Res + r_n^2}\right)\left[\frac{b_0}{s} + \frac{b_1}{s^2+\omega^2} + A_0\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{Re^2} + \frac{Gm\phi_H(r,s)}{Re^2}$$

$$u_H(r_n,s) = \left(\frac{1}{s + \frac{r_n^2}{Re}}\right)\left[\frac{b_0}{s} + \frac{b_1}{s^2+\omega^2} + A_0\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{Re^2} + \frac{Gm\phi_H(r,s)}{Re^2}$$

$$u_H(r_n,s) = \left(\frac{1}{s+\eta}\right)\left[\frac{b_0}{s} + \frac{b_1}{s^2+\omega^2} + A_0\left(\frac{s\cos(\phi_0) - k\sin(\phi_0)}{s^2+k^2}\right)\right]\frac{J_1 r_n}{r_n} + \frac{Gr\theta_H(r,s)}{Re^2} + \frac{Gm\phi_H(r,s)}{Re^2} \tag{3.2.4}$$

$$(sPe-R)\theta_H(r_n,s) = -r_n^2\phi_H(r_n,s) + \frac{Pe(Q_m+\theta_m)}{s}\frac{J_1(r_n)}{r_n}$$

$$(sPe-R)\theta_H(r_n,s) + r_n^2\phi_H(r_n,s) = \frac{Pe(Q_m+\theta_m)}{s}\frac{J_1(r_n)}{r_n}$$

$$(sPe-R+r_n^2)\theta_H(r_n,s) = \frac{Pe(Q_m+\theta_m)}{s}\frac{J_1(r_n)}{r_n}$$

$$\theta_H(r_n,s) = \frac{Pe(Q_m+\theta_m)}{Pes(s+\frac{r_n^2-Re}{Pe})}\frac{J_1(r_n)}{r_n}$$

$$\theta_H(r_n,s) = \frac{(Q_m+\theta_m)}{s(s+\delta)}\frac{J_1(r_n)}{r_n} \tag{3.2.5}$$

$$sReSc\phi_H(r_n,s) = -r_n^2\phi_H(r_n,s) + SrSc(-r_n^2\theta_H(r_n,s))$$

$$sReSc\phi_H(r_n,s) + r_n^2\phi_H(r_n,s) = -SrScr_n^2\left(\frac{(Q_m+\theta_m)}{s(s+\delta)}\frac{J_1(r_n)}{r_n}\right)$$

$$ReSc(s+\frac{r_n^2}{ReSc})\phi_H(r_n,s) = -SrScr_n^2\left(\frac{(Q_m+\theta_m)}{s(s+\delta)}\frac{J_1(r_n)}{r_n}\right)$$

$$\phi_H(r_n,s) = -\frac{r_n^2Sr}{Re(s+\frac{r_n^2}{ReSc})}\left(\frac{(Q_m+\theta_m)}{s(s+\delta)}\frac{J_1(r_n)}{r_n}\right)$$

$$\phi_H(r_n,s) = -\frac{r_n^2Sr}{Re(s+\lambda)}\left(\frac{(Q_m+\theta_m)}{s(s+\delta)}\frac{J_1(r_n)}{r_n}\right) \tag{3.2.6}$$

Where $\eta = \frac{r_n^2}{Re}$,$\delta = \frac{r_n^2-Re}{Pe}$ and $\lambda = \frac{r_n^2}{ReSc}$

By swapping the values of $\theta_H(r_n,s)$ and $\phi_H(r_n,s)$ in equation (3.2.4) then we get,

$$u_H(r_n,s) = \left[\frac{b_0}{s(s+\eta)} + \frac{b_1}{(s^2+\omega^2)(s+\eta)} + A_0\left(\frac{s\cos(\phi_0)-k\sin(\phi_0)}{(s^2+k^2)(s+\eta)}\right)\right]\frac{J_1r_n}{r_n} \tag{3.2.7}$$

$$+\frac{Gr}{Re^2}\left(\frac{(Q_m+\theta_m)}{s(s+\eta)(s+\delta)}\right)\frac{J_1(r_n)}{r_n} + \frac{Gm}{Re^2}\left(-\frac{r_n^2Sr}{Re}\frac{(Q_m+\theta_m)}{s(s+\eta)(s+\delta)(s+\lambda)}\right)\frac{J_1(r_n)}{r_n}$$

Next, we will utilize the inverse Laplace transform on the given equations.(3.2.5) – (3.2.7),

$$\mathcal{L}^{-1}\left(u_H(r_n,s)\right)=\mathcal{L}^{-1}\left[\frac{b_0}{s(s+\eta)}+\frac{b_1}{(s^2+\omega^2)(s+\eta)}+A_0\left(\frac{s\cos(\phi_0)-k\sin(\phi_0)}{(s^2+k^2)(s+\eta)}\right)\right]\frac{J_1r_n}{r_n}$$
$$+\mathcal{L}^{-1}\left[\frac{Gr}{Re^2}\left(\frac{(Q_m+\theta_m)}{s(s+\eta)(s+\delta)}\right)\frac{J_1(r_n)}{r_n}\right]+\mathcal{L}^{-1}\left[\frac{Gm}{Re^2}\left(-\frac{r_n^2Sr}{Re}\frac{(Q_m+\theta_m)}{s(s+\eta)(s+\delta)(s+\lambda)}\right)\frac{J_1(r_n)}{r_n}\right]$$

$$\begin{aligned}\mathcal{L}^{-1}\left(u_H(r_n,s)\right)=&\mathcal{L}^{-1}\left[b_0\frac{1}{s(s+\eta)}+b_1\frac{1}{(s^2+\omega^2)(s+\eta)}+A_0\left(\frac{s\cos(\phi_0)-k\sin(\phi_0)}{(s^2+k^2)(s+\eta)}\right)\right]\frac{J_1r_n}{r_n}\\&+\frac{Gr(Q_m+\theta_m)}{Re^2}\mathcal{L}^{-1}\left[\left(\frac{1}{s(s+\eta)(s+\delta)}\right)\frac{J_1(r_n)}{r_n}\right]\\&-\frac{Gmr_n^2Sr(Q_m+\theta_m)}{Re^3}\mathcal{L}^{-1}\left[\left(\frac{1}{s(s+\eta)(s+\delta)(s+\lambda)}\right)\frac{J_1(r_n)}{r_n}\right]\end{aligned}$$

$$\begin{aligned}u_H(r_n,t)=&\left[b_0\left(\frac{1-e^{-\eta t}}{\eta}\right)+b_1\left(\frac{e^{-\eta t}}{\eta}+\frac{\eta\sin t\omega-\omega\cos t\omega}{\omega(\eta^2+\omega^2)}\right)\right]\frac{J_1r_n}{r_n}\qquad(3.2.8)\\&+A_0\left(-\frac{e^{-\eta(t)}(\eta\cos(\phi)+k\sin(\phi))}{\eta^2+k^2}\right)\frac{J_1r_n}{r_n}\\&+A_0\left(\frac{\eta\cos(\phi)\cos(kt)+k\cos(\phi)\sin(kt)+k\sin(\phi)\cos(kt)-\eta\sin(\phi)\sin(kt)}{\eta^2+k^2}\right)\frac{J_1r_n}{r_n}\\&+\frac{Gr(Q_m+\theta_m)}{Re^2}\left(\frac{1}{\eta\delta}+\frac{e^{-\delta t}}{\delta(\delta-\eta)}-\frac{e^{-\eta t}}{\eta(\delta-\eta)}\right)\frac{J_1(r_n)}{r_n}\\&-\frac{Gmr_n^2Sr(Q_m+\theta_m)}{Re^3}\left(\frac{1}{\eta\delta\lambda}+\frac{e^{-\eta t}}{\delta(\delta-\eta)(\eta-\lambda)}+\frac{e^{-\lambda t}}{\lambda(\delta-\lambda)(\lambda-\eta)}-\frac{e^{-\delta t}}{\delta(\delta-\eta)(\delta-\lambda)}\right)\frac{J_1(r_n)}{r_n}\end{aligned}$$

$$\mathcal{L}^{-1}(\theta_H(r_n,s))=(Q_m+\theta_m)\mathcal{L}^{-1}\left(\frac{1}{s(s+\delta)}\right)\frac{J_1(r_n)}{r_n}$$

$$\theta_H(r_n,t)=(Q_m+\theta_m)\left(\frac{1-e^{-\delta t}}{\delta}\right)\frac{J_1(r_n)}{r_n}\qquad(3.2.9)$$

$$\mathcal{L}^{-1}(\phi_H(r_n,s))=-\frac{r_n^2Sr(Q_m+\theta_m)}{Re}\mathcal{L}^{-1}\left(\frac{1}{s(s+\delta)(s+\lambda)}\right)\frac{J_1(r_n)}{r_n}$$

$$\phi_H(r_n,t)=-\frac{r_n^2Sr(Q_m+\theta_m)}{Re}\left(\frac{1}{\delta\lambda}+\frac{e^{-\delta(t)}}{\delta(\delta-\lambda)}-\frac{e^{-\lambda t}}{\eta(\delta-\eta)}\right)\frac{J_1(r_n)}{r_n}\qquad(3.2.10)$$

The equation is further solved using the inverse Hankel transform (3.2.8) − (3.2.10), we get

$$
\begin{aligned}
u(r,t) =& 2\pi \sum_{n=1}^{\infty} \frac{J_0(r_n r)}{r_n J_1(r_n)} b_0 \left(\frac{1-e^{-\eta t}}{\eta}\right) + b_1 \left(\frac{e^{-\eta t}}{\eta} + \frac{\eta \sin t\omega - \omega \cos t\omega}{\omega(\eta^2+\omega^2)}\right) \qquad (3.2.11)\\
&+ A_0 \left(-\frac{e^{-\eta t}(\eta \cos(\phi) + k \sin(\phi))}{\eta^2+k^2}\right)\\
&+ A_0 \left(\frac{\eta \cos(\phi)\cos(kt) + k\cos(\phi)\sin(kt) + k\sin(\phi)\cos(kt) - \eta\sin(\phi)\sin(kt)}{\eta^2+k^2}\right)\\
&+ \frac{Gr(Q_m+\theta_m)}{Re^2}\left(\frac{1}{\eta\delta} + \frac{e^{-\delta t}}{\delta(\delta-\eta)} - \frac{e^{-\eta t}}{\eta(\delta-\eta)}\right)\\
&- \frac{Gmr_n^2 Sr(Q_m+\theta_m)}{Re^3}\left(\frac{1}{\eta\delta\lambda} + \frac{e^{-\eta t}}{\delta(\delta-\eta)(\eta-\lambda)} + \frac{e^{-\lambda t}}{\lambda(\delta-\lambda)(\lambda-\eta)} - \frac{e^{-\delta t}}{\delta(\delta-\eta)(\delta-\lambda)}\right)
\end{aligned}
$$

$$
\theta(r,t) = 2\pi \sum_{n=1}^{\infty} \frac{J_0(r_n r)}{r_n J_1(r_n)} \left((Q_m+\theta_m)\left(\frac{1-e^{-\delta t}}{\delta}\right)\right) \qquad (3.2.12)
$$

$$
\phi(r,t) = -2\pi \sum_{n=1}^{\infty} \frac{J_0(r_n r)}{r_n J_1(r_n)} \left(\frac{r_n^2 Sr(Q_m+\theta_m)}{Re}\left(\frac{1}{\delta\lambda} + \frac{e^{-\delta t}}{\delta(\delta-\lambda)} - \frac{e^{-\lambda t}}{\eta(\delta-\eta)}\right)\right) \qquad (3.2.13)
$$

Chapter 4

Results and Discussions

To investigate the influence of different variables on velocity, temperature, and coolant concentration under conditions of acceleration, this section includes graphical data presentation. Utilizing solved equations, we have developed a MathCAD code capable of generating various graphs. These graphs depict the impact of radiation on several non-dimensional parameters such as Thermal Grashof number Gr, Solutal Grashof number Gm,Reynolds number Re, Schmidt number Sc, Prandtl number Pr, Darcy's number Da, Radiation number R, etc.

The examination encompassed an analysis of the coolant's velocity, temperature, and concentration.Certain parameter values were held constant to facilitate numerical computations that are $A_0 = 0.05$, $b_0 = 0.5$, $b_1 = 0.5$, $\phi = \frac{\pi}{4}$, $\omega = \frac{\pi}{4}$,$R = 1$, $Pe = 15000$, $Q_m = 100$, $Sr = 1.5$, $Sc = 1.5$, $Re = 4$, $Da = 0.15$, $Gr = 1200$, $Gm = 1500$, $\eta = 0.15$, $\lambda = 0.2$, $\delta = 0.3$.The relationship between the velocity profile and heat absorption of glycols is evident when examining the figures 4.1, 4.2, and 4.3. This suggests that as the heat absorption rate increases, the engine achieves a more rapid and stable state.

As Ethylene Glycol possesses exceptional thermal properties with a heat absorption ($\theta_m = 2.42$), making it highly efficient in dissipating heat from the engine and ensuring temperature stability. In comparison, propylene glycol and diethylene glycol have lower heat absorptions, with values of ($\theta_m = 2.21$) and ($\theta_m = 2.03$), respectively. Figure (4.4) illustrates that nanofluids containing ethylene glycol reach temperature stability rapidly, even at elevated temperatures, due to their elevated heat capacity. Conversely, nanofluids containing propylene glycol and diethylene glycol stabilize at lower temperatures, as depicted in Figures (4.5) and (4.6), respectively.

In figures 4.7,4.8, and 4.9, it is evident that there exists a reverse correlation between heat absorption (θ_m) and concentration. This suggests that as the value of (θ_m) decreases, the concentration profile undergoes rapid stabilization.

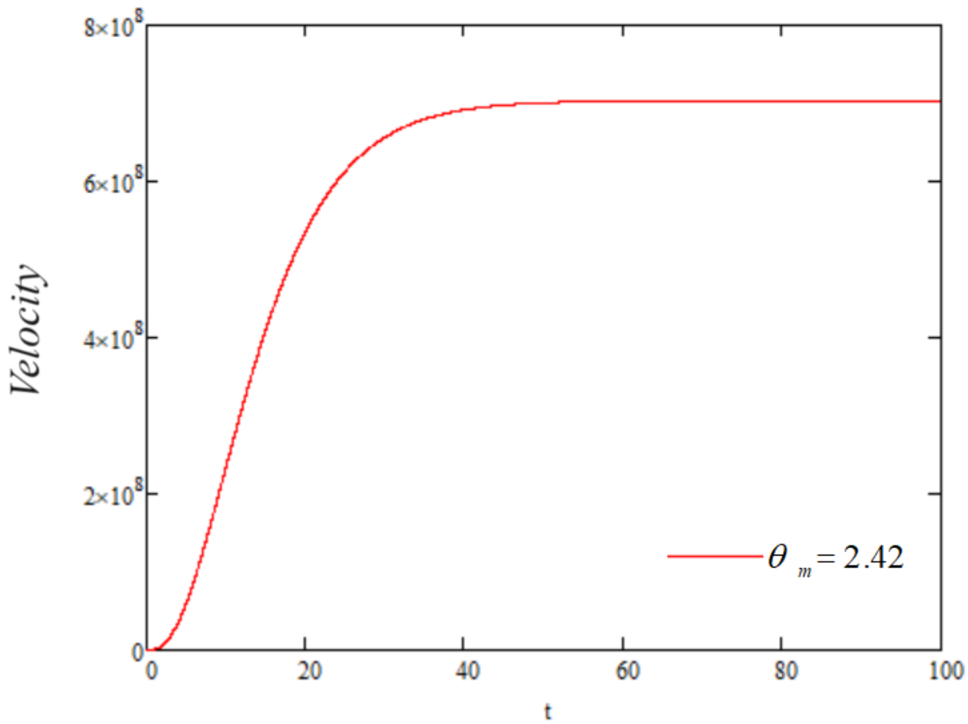

Figure 4.1: Change in velocity over time for Ethylene Glycol

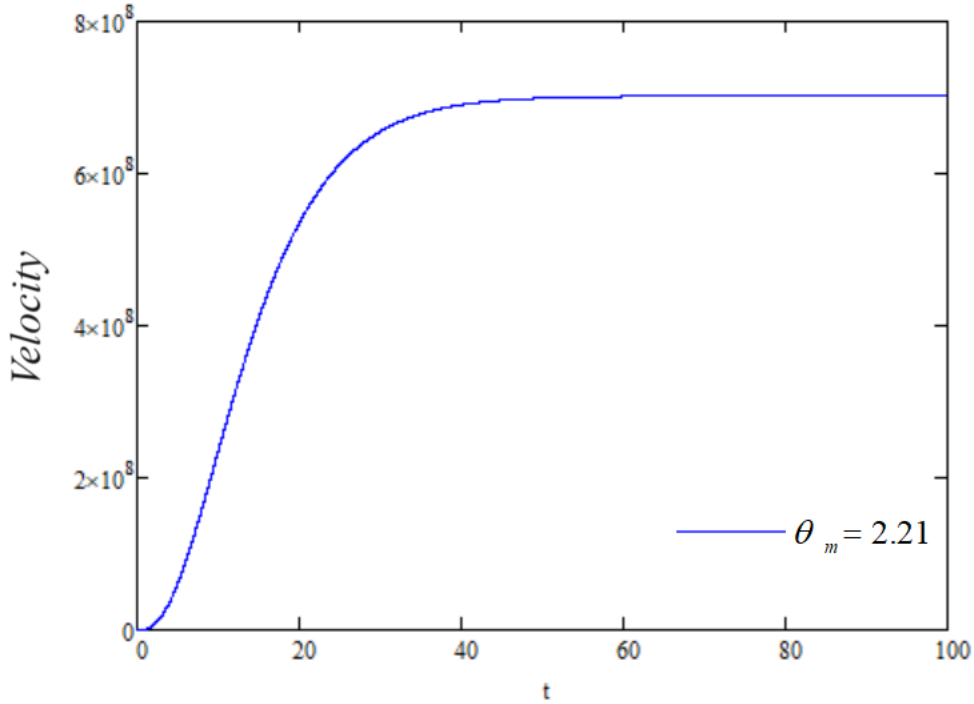

Figure 4.2: Change in velocity over time for Propylene Glycol

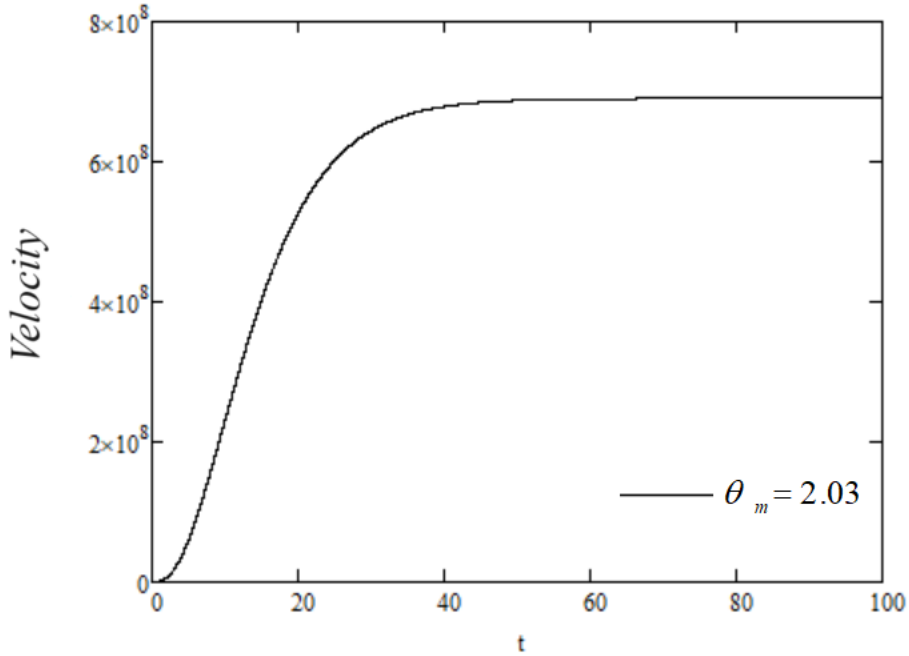

Figure 4.3: Change in velocity over time for Diethylene Glycol

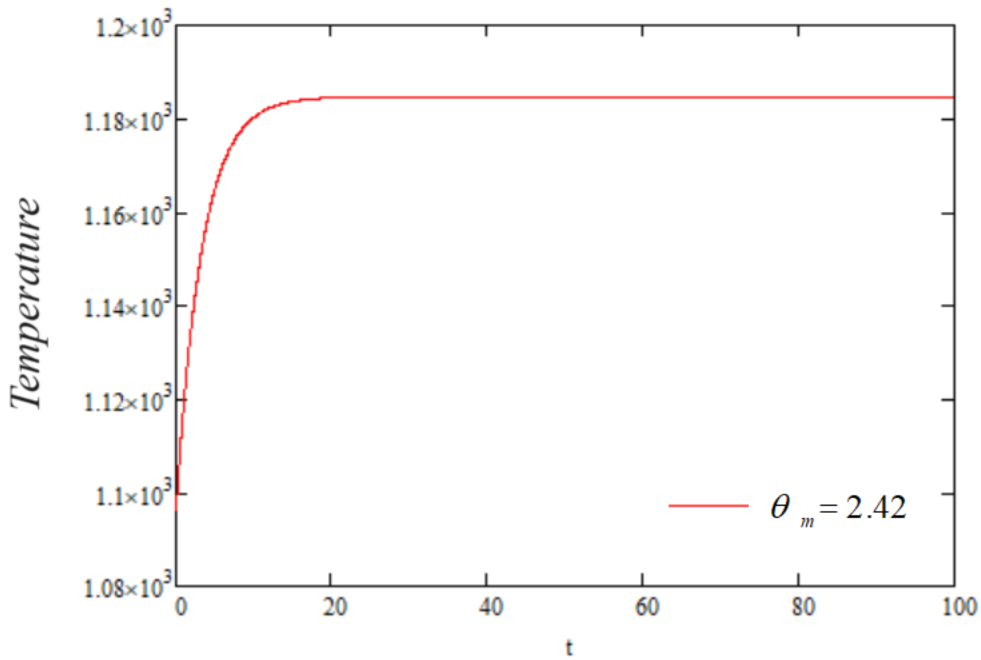

Figure 4.4: Variation of temperature versus t for Ethylene Glycol

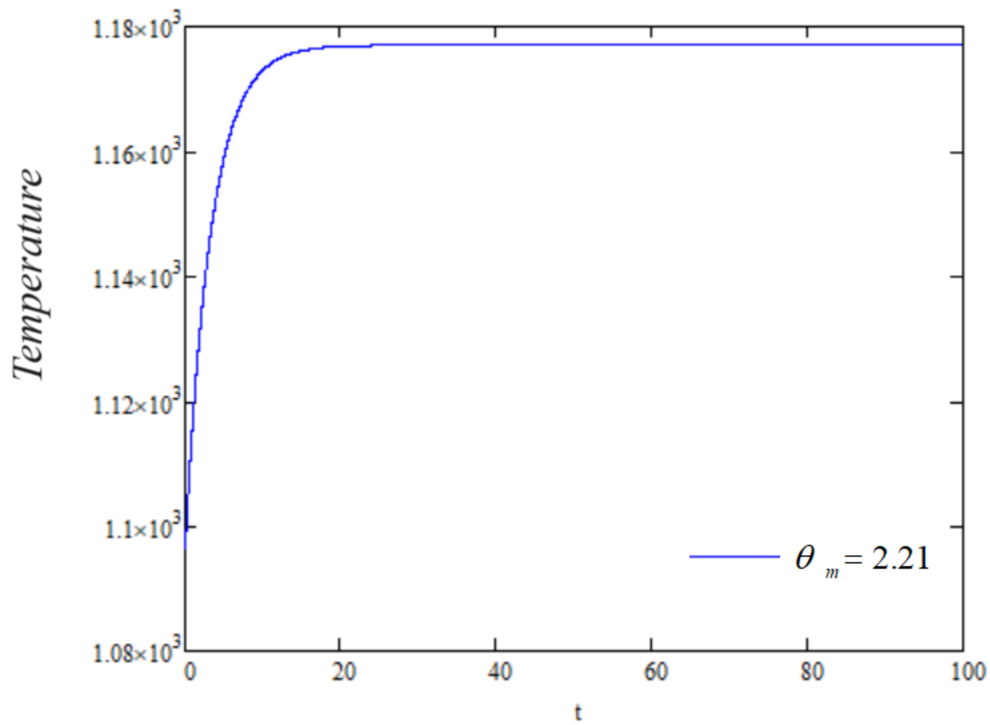

Figure 4.5: Variation of temperature versus t for Propylene Glycol

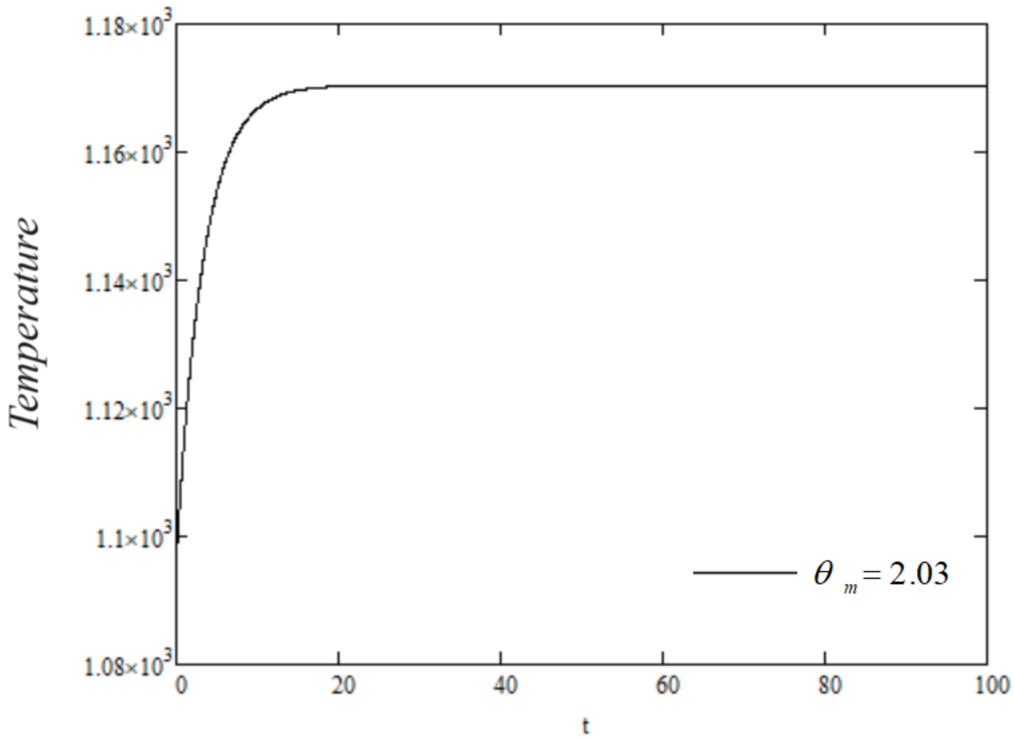

Figure 4.6: Variation of temperature versus t for Diethylene Glycol

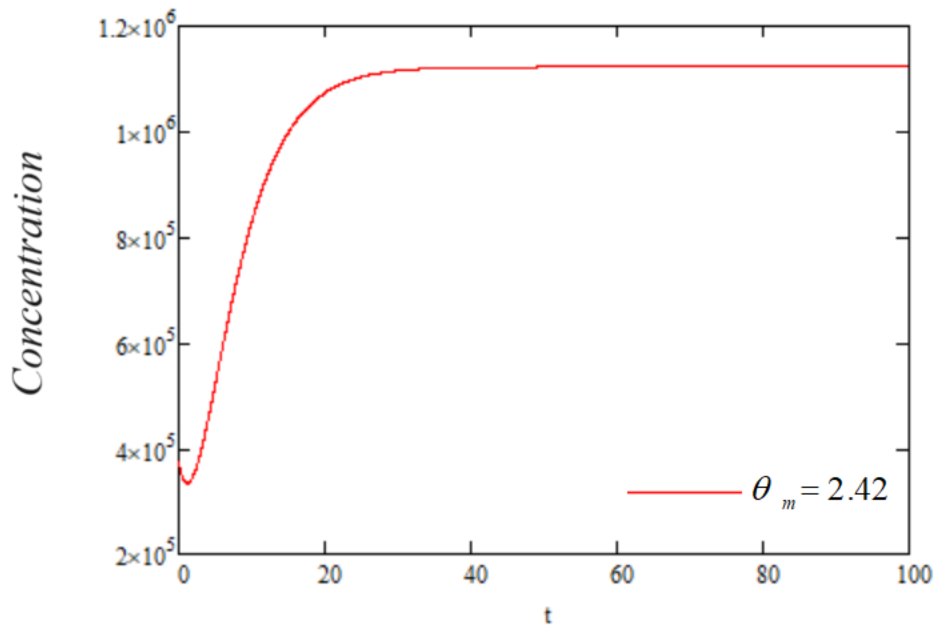

Figure 4.7: Concentration variation versus time for Ethylene Glycol

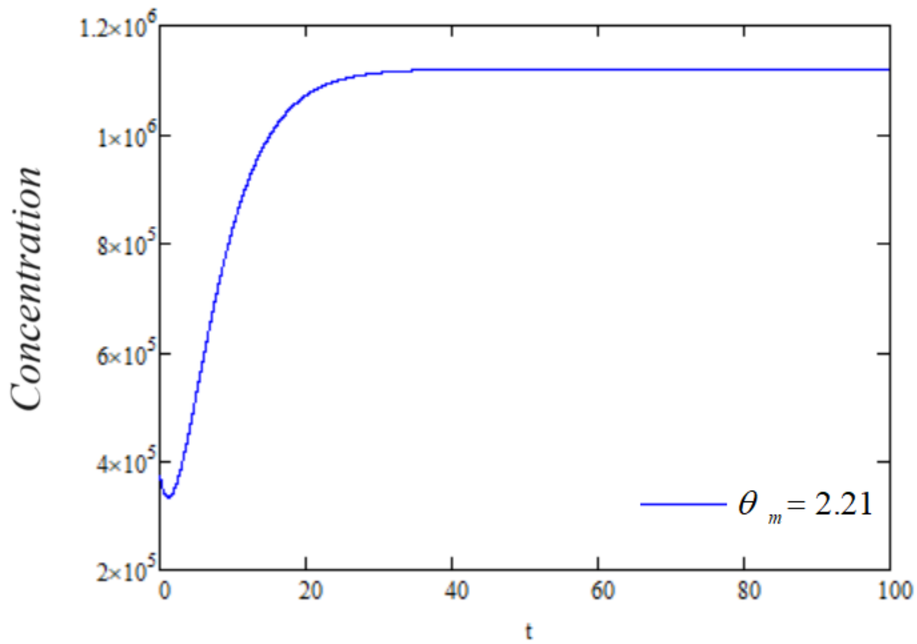

Figure 4.8: Concentration variation versus time for Proplyene Glycol

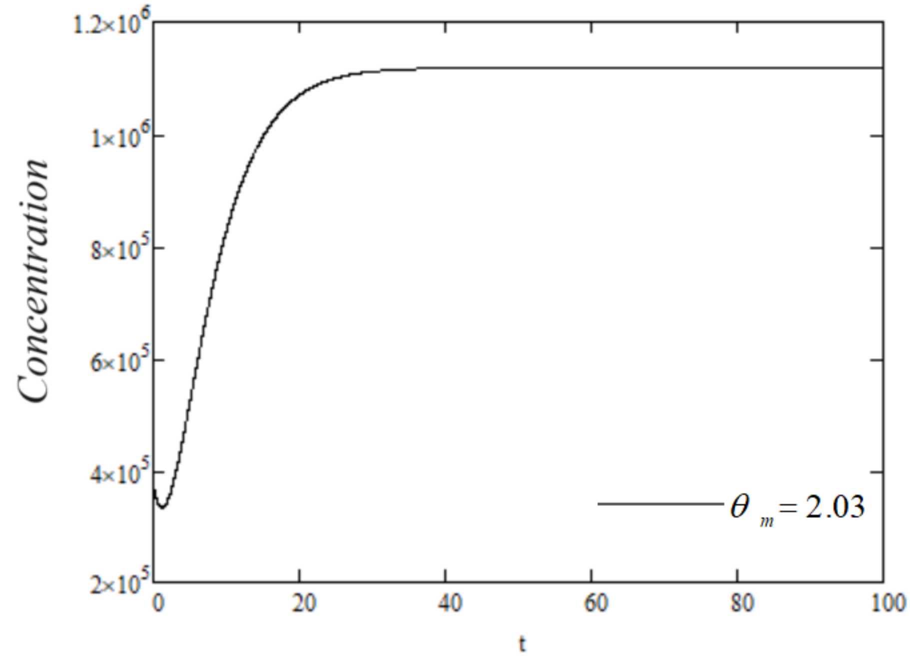

Figure 4.9: Concentration variation versus time for Diethylene Glycol

Chapter 5

Conclusion

In this thesis, our focus revolves around the examination of coolant flow within cylindrical tubes found in an automotive radiator. To delve into this subject, we have employed a mathematical model that comprises coupled differential equations governing temperature, concentration, and coolant velocity. Our approach to solving this model involves the utilization of Laplace and finite Hankel transforms. The outcome of this endeavour is the derivation of analytical expressions for coolant flow velocity, temperature, and concentration within the radiator.In summary, ethylene glycol emerges as the preferred choice for car coolant when compared to propylene and diethylene glycols. Its exceptional heat transfer capabilities ensure efficient heat dissipation from the engine, maintaining ideal operating temperatures. Moreover, its lower freezing point provides superior protection against freezing, especially in cold weather conditions. Additionally, ethylene glycol-based coolants come equipped with effective corrosion inhibitors, safeguarding critical components like the radiator and water pump and extending their lifespan. In essence, ethylene glycol excels in thermal efficiency, anti-freezing properties, and corrosion protection, making it the optimal selection for automotive cooling systems.

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